



# Rafael Tiedra

*Mathematician/Data Scientist with various years of experience in research, IT, education and administration.*

Date of birth 05.03.1976

Nationality Swiss and Spanish

Languages French (native), English and Spanish (fluent), German (beginner)

## Skills

### Research and teaching

Problem-solving, research in physics and mathematics (40+ scientific publications)

Scientific communication and vulgarisation (150+ talks)

Classroom and online teaching

### Academic and corporate administration

Management, budgets, fundraising, organisation of events and conferences

Writing and editing publications, popular science, translation, digital communication

Recruitment process, conception of curriculums, project management

## Programming

**Languages/tools:** Python, SQL, TensorFlow, PyTorch, Scikit-Learn, VSCode, Jupyter, Git, OpenShift, Qiskit, CP-SAT, HTML, LaTeX

**IT:** MS Office, Linux, Windows OS

## Professional experiences

- Aug. 2022-present **Data Scientist**, Department of Data Science and AI, Geneva State, Switzerland
- Aug. 2021-July 2022 **Visiting professor**, Swiss Federal Institute of Technology Lausanne, Switzerland
- Jan. 2009-Feb. 2022 **Professor**, Pontifical Catholic University, Chile
- Oct. 2005-Dec. 2008 **Postdoctoral researcher**, Universities of Paris-Sud and Cergy-Pontoise, France

## Studies

- June 2005 **PhD in Physics**, University of Geneva
- Oct. 2001 **BSc in Mathematics**, University of Geneva
- June 2001 **MSc in Physics**, University of Geneva

## Prizes and research grants

- 2009, 2013, 2017, 2021 Chilean Fondecyt grants (501'000 USD)
- 2009, 2014 Chilean Milenio grants (2'338'000 USD)
- 2011 French-Chilean ECOS-CONICYT grant
- 2005, 2007 Swiss NSF grants (138'000 USD)
- 2001 Fanny Wurth scholarship
- 2001 Hans Wilsdorf scholarship (Rolex)
- 1998, 1999, 2000, 2001 Canton of Geneva scholarships